

**B10 Partial fractions A level
only**

Name: _____

Class: _____

Date: _____

Time: **237 minutes**

Marks: **197 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

Find the value of $\int_1^2 \frac{6x+1}{6x^2-7x+2} dx$, expressing your answer in the form $m \ln 2 + n \ln 3$, where m and n are integers.

(Total 8 marks)

Q2.

It is given that $f(x) = \frac{7x-1}{(1+3x)(3-x)}$

- (a) Express $f(x)$ in the form $\frac{A}{3-x} + \frac{B}{1+3x}$, where A and B are integers. (3)
- (b) (i) Find the first three terms of the binomial expansion of $f(x)$ in the form $a + bx + cx^2$, where a , b and c are rational numbers. (7)
- (ii) State why the binomial expansion cannot be expected to give a good approximation to $f(x)$ at $x = 0.4$. (1)

(Total 11 marks)

Q3.

- (a) (i) Express $\frac{5-8x}{(2+x)(1-3x)}$ in the form $\frac{A}{2+x} + \frac{B}{1-3x}$, where A and B are integers. (3)

- (ii) Hence show that $\int_{-1}^0 \frac{5-8x}{(2+x)(1-3x)} dx = p \ln 2$, where p is rational. (4)

- (b) (i) Given that $\frac{9-18x-6x^2}{2-5x-3x^2}$ can be written as $C + \frac{5-8x}{2-5x-3x^2}$, find the value of C . (1)

- (ii) Hence find the exact value of the area of the region bounded by the curve $y = \frac{9-18x-6x^2}{2-5x-3x^2}$, the x -axis and the lines $x = -1$ and $x = 0$.

You may assume that $y > 0$ when $-1 \leq x \leq 0$.

(2)

(Total 10 marks)

Q4.

- (a) Express $\frac{2x+3}{4x^2-1}$ in the form $\frac{A}{2x-1} + \frac{B}{2x+1}$, where A and B are integers. (3)

- (b) Express $\frac{12x^3-7x-6}{4x^2-1}$ in the form $Cx + \frac{D(2x+3)}{4x^2-1}$, where C and D are integers. (3)

- (c) Evaluate $\int_1^2 \frac{12x^3-7x-6}{4x^2-1} dx$, giving your answer in the form $p + \ln q$, where p and q are rational numbers. (5)

(Total 11 marks)

Q5.

- (a) (i) Express $\frac{5x-6}{x(x-3)}$ in the form $\frac{A}{x} + \frac{B}{x-3}$. (2)

- (ii) Find $\int \frac{5x-6}{x(x-3)} dx$. (2)

- (b) (i) Given that

$$4x^3 + 5x - 2 = (2x+1)(2x^2 + px + q) + r$$

find the values of the constants p , q and r . (4)

- (ii) Find $\int \frac{4x^3+5x-2}{2x+1} dx$. (3)

(Total 11 marks)

Q6.

- (a) Express $\frac{3+9x}{(1+x)(3+5x)}$ in the form $\frac{A}{1+x} + \frac{B}{3+5x}$, where A and B are integers. (3)

- (b) Hence, or otherwise, find the binomial expansion of $\frac{3+9x}{(1+x)(3+5x)}$ up to and including the term in x^2 . (7)
- (c) Find the range of values of x for which the binomial expansion of $\frac{3+9x}{(1+x)(3+5x)}$ is valid. (2)
- (Total 12 marks)

Q7.

- (a) Express $\frac{1}{(3-2x)(1-x)^2}$ in the form $\frac{A}{3-2x} + \frac{B}{1-x} + \frac{C}{(1-x)^2}$. (4)

- (b) Solve the differential equation

$$\frac{dy}{dx} = \frac{2\sqrt{y}}{(3-2x)(1-x)^2}$$

where $y = 0$ when $x = 0$, expressing your answer in the form

$$y^p = q \ln[f(x)] + \frac{x}{1-x}$$

where p and q are constants.

(9)

(Total 13 marks)

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Q8.

The expression $\frac{10x^2+8}{(x+1)(5x-1)}$ can be written in the form $2 + \frac{A}{x+1} + \frac{B}{5x-1}$, where A and B are constants.

- (a) Find the values of A and B . (4)

- (b) Hence find $\int \frac{10x^2+8}{(x+1)(5x-1)} dx$.

(4)

(Total 8 marks)

Q9.

(a) (i) Express $\frac{7x-3}{(x+1)(3x-2)}$ in the form $\frac{A}{x+1} + \frac{B}{3x-2}$. (3)

(ii) Hence find $\int \frac{7x-3}{(x+1)(3x-2)} dx$. (2)

(b) Express $\frac{6x^2 + x + 2}{2x^2 - x + 1}$ in the form $P + \frac{Qx+R}{2x^2 - x + 1}$. (3)

(Total 8 marks)

Q10.

(a) (i) Express $\frac{2x+7}{x+2}$ in the form $A + \frac{B}{x+2}$, where A and B are integers. (2)

(ii) Hence find $\int \frac{2x+7}{x+2} dx$. (2)

(b) (i) Express $\frac{28+4x^2}{(1+3x)(5-x)^2}$ in the form $\frac{P}{1+3x} + \frac{Q}{5-x} + \frac{R}{(5-x)^2}$, where P , Q and R are constants. (5)

(ii) Hence find $\int \frac{28+4x^2}{(1+3x)(5-x)^2} dx$. (4)

(Total 13 marks)

Q11.

Express $\frac{3x^3 + 8x^2 - 3x - 5}{3x-1}$ in the form $ax^2 + bx + \frac{c}{3x-1}$, where a , b and c are integers. (Total 3 marks)

Q12.

(a) Find the binomial expansion of $(1-x)^{-1}$ up to and including the term in x^2 . (2)

- (b) (i) Express $\frac{3x-1}{(1-x)(2-3x)}$ in the form $\frac{A}{1-x} + \frac{B}{2-3x}$, where A and B are integers. (3)
- (ii) Find the binomial expansion of $\frac{3x-1}{(1-x)(2-3x)}$ up to and including the term in x^2 . (6)
- (c) Find the range of values of x for which the binomial expansion of $\frac{3x-1}{(1-x)(2-3x)}$ is valid. (2)
- (Total 13 marks)**

Q13.

- (a) Given that $\frac{3}{9-x^2}$ can be expressed in the form $k\left(\frac{1}{3+x} + \frac{1}{3-x}\right)$, find the value of the rational number k . (2)
- (b) Show that $\int_1^2 \frac{3}{9-x^2} dx = \frac{1}{2} \ln\left(\frac{a}{b}\right)$, where a and b are integers. (3)
- (Total 5 marks)**

Q14.

- (a) The polynomial $f(x)$ is defined by $f(x) = 2x^3 + 3x^2 - 18x + 8$.
- (i) Use the Factor Theorem to show that $(2x - 1)$ is a factor of $f(x)$. (2)
- (ii) Write $f(x)$ in the form $(2x - 1)(x^2 + px + q)$, where p and q are integers. (2)
- (iii) Simplify the algebraic fraction $\frac{4x^2 + 16x}{2x^3 + 3x^2 - 18x + 8}$. (2)
- (b) Express the algebraic fraction $\frac{2x^2}{(x+5)(x-3)}$ in the form $A + \frac{B+Cx}{(x+5)(x-3)}$, where A , B and C are integers. (4)

(Total 10 marks)

Q15.

- (a) Express $\frac{2}{x^2 - 1}$ in the form $\frac{A}{x-1} + \frac{B}{x+1}$. (3)

- (b) Hence find $\int \frac{2}{x^2 - 1} dx$. (2)

- (c) Solve the differential equation $\frac{dy}{dx} = \frac{2y}{3(x^2 - 1)}$, given that $y = 1$ when $x = 3$.

Show that the solution can be written as $y^3 = \frac{2(x-1)}{x+1}$. (5)

(Total 10 marks)

Q16.

- (a) (i) Express $\frac{3x-5}{x-3}$ in the form $A + \frac{B}{x-3}$, where A and B are integers. (2)

- (ii) Hence find $\int \frac{3x-5}{x-3} dx$. (2)

- (b) (i) Express $\frac{6x-5}{4x^2-25}$ in the form $\frac{P}{2x+5} + \frac{Q}{2x-5}$, where P and Q are integers. (3)

- (ii) Hence find $\int \frac{6x-5}{4x^2-25} dx$. (3)

(Total 10 marks)

Q17.

- (a) (i) Find the binomial expansion of $(1+x)^{-1}$ up to the term in x^3 . (2)

- (ii) Hence, or otherwise, obtain the binomial expansion of $\frac{1}{1+3x}$ up to the term in x^3 . (2)

- (b) Express $\frac{1+4x}{(1+x)(1+3x)}$ in partial fractions. (3)

- (c) (i) Find the binomial expansion of $\frac{1+4x}{(1+x)(1+3x)}$ up to the term in x^3 . (3)

- (ii) Find the range of values of x for which the binomial expansion of $\frac{1+4x}{(1+x)(1+3x)}$ is valid. (2)

(Total 12 marks)

Q18.

- (a) (i) Obtain the binomial expansion of $(1-x)^{-1}$ up to and including the term in x^2 . (2)
- (ii) Hence, or otherwise, show that

$$\frac{1}{3-2x} \approx \frac{1}{3} + \frac{2}{9}x + \frac{4}{27}x^2$$

for small values of x .

- (b) Obtain the binomial expansion of $\frac{1}{(1-x)^2}$ up to and including the term in x^2 . (2)

- (c) Given that $\frac{2x^2-3}{(3-2x)(1-x)^2}$ can be written in the form $\frac{A}{(3-2x)} + \frac{B}{(1-x)} + \frac{C}{(1-x)^2}$, find the values of A , B and C . (5)

- (d) Hence find the binomial expansion of $\frac{2x^2-3}{(3-2x)(1-x)^2}$ up to and including the term in x^2 . (3)

(Total 15 marks)

Q19.

- (a) Given that $\frac{9x^2 - 6x + 5}{(3x-1)(x-1)}$ can be written in the form $3 + \frac{A}{3x-1} + \frac{B}{x-1}$, where A and B are integers, find the values of A and B .

(4)

- (b) Hence, or otherwise, find $\int \frac{9x^2 - 6x + 5}{(3x-1)(x-1)} dx$.

(4)

(Total 8 marks)

Q20.

- (a) Express $\frac{3x-5}{(x+3)(2x-1)}$ in the form $\frac{A}{x+3} + \frac{B}{2x-1}$.

(3)

- (b) Hence find $\int \frac{3x-5}{(x+3)(2x-1)} dx$.

(3)

(Total 6 marks)

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